

THE CAPITOL SPORTSBOOK

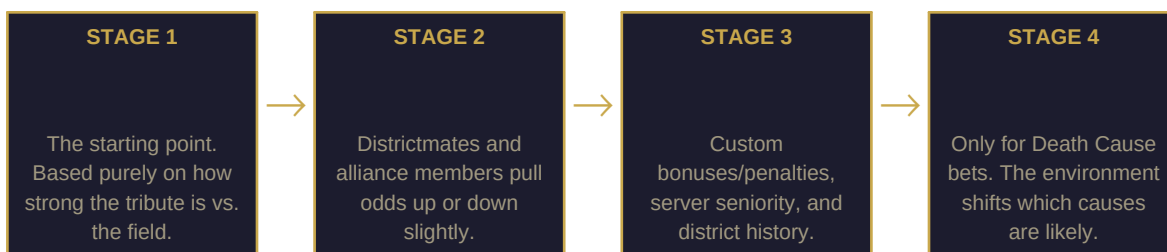
How Tribute Odds Work

A Plain-Language Guide for Bookmakers & Spectators

Every number you see next to a tribute's name tells a story. This guide explains exactly what shapes those numbers — from training scores and alliances to server history and the arena itself.

How the Odds Pipeline Works

Every time a tribute enters or leaves the arena, all open markets are recalculated automatically. The odds travel through four stages — each one layering additional context on top of the last.



Markets with a manual odds override are frozen — they skip all four stages and never change unless you update them by hand.

Stage 1 — Training Score (The Starting Point)

Every tribute's odds begin with one question: **how does their training score compare to everyone still alive?** A tribute with a score of 10 in a field where the total is 100 starts with a 10% chance of winning. The math below shows how that percentage is adjusted per market type.

Market	What Drives the Odds	Key Detail
Victor	Tribute's score as a % of all alive scores	1 point = 1% more likely to win in a 100-point field
Exact Placement	Win % × a placement multiplier	2nd place is ×1.6; 5th is ×3.5; the further back, the bigger the boost
Top-N Finish	Win % × the N value × 0.7	"Top 4" gives a ×2.8 boost before dampening
Top Killer	Score squared vs. field's scores squared	Score gaps are amplified — a 10 vs. 8s is a much bigger edge here than in Victor
Kill Event (A→B)	Both tributes' strengths × field size	Bigger fields = more chances for them to meet
Bloodbath Survivor	Inverted — weaker tributes start as favorites	Top scorers attract attention; weaker tributes slip past unnoticed
First Blood	Same as Victor market	Proportional score share
Kills Over/Under	Expected kill rate from score share	Higher score = higher expected kills; line adjusts threshold
Placement Over/Under	Score share × the line value	Under = finishes better than line; Over = finishes worse
Death Cause	Starts flat at +200	Adjusted only by the Arena Factor (Stage 4)

Market	What Drives the Odds	Key Detail
Custom Markets	Difficulty setting or manual odds	Easy ≈ -200 · Moderate $\approx +100$ · Hard $\approx +300$ · Very Hard $\approx +700$ · Longshot $\approx +1500$

Exact Placement Multipliers

The further back the placement, the larger the multiplier applied to the tribute's win percentage — reflecting how much easier it is to finish 8th than 1st.

Placement	Multiplier	Example — tribute at 10% win odds
2nd	$\times 1.6$	16% $\rightarrow$ approx. -95
3rd	$\times 2.2$	22% $\rightarrow$ approx. -38
4th	$\times 2.8$	28% $\rightarrow$ approx. -39
5th	$\times 3.5$	35% $\rightarrow$ approx. -54
6th	$\times 4.0$	40% $\rightarrow$ approx. -67
7th	$\times 4.5$	45% $\rightarrow$ approx. -82
8th	$\times 5.0$	50% $\rightarrow -100$ (even money)
9th+	$\times (\text{place} \times 0.8)$	Scales linearly beyond 8th

Stage 2 — Peer Influence (District & Alliance)

After the base odds are set, the system looks at who else is still alive from the same district and the same alliance. Tributes are subtly pulled toward their peers — if your partner is a favourite, you become a slightly bigger favourite too. If they're an underdog, you're nudged downward.

Peer Group	Influence Strength	Who Counts
District	10% pull toward the peer average	Living tributes from the same district only
Alliance	20% pull toward the peer average	Living tributes in the same named alliance only

What This Means in Practice

- A weak tribute allied with strong tributes will see their odds slightly improve.
- A strong tribute in a weak district will have their odds very slightly dragged down.
- Alliance influence is twice as strong as district influence.
- Only alive tributes contribute — dead peers no longer affect anyone.
- This stage has no effect on Kill Event, Death Cause, or custom market types.

Stage 3 — The Modifier Factor

This is where admin-controlled context gets layered in. Four things combine into a single multiplier that nudges the probability up or down before the final odds are calculated.

Layer A — Custom Odds Modifiers

These are reusable multipliers you create via **/admin modifier**. You assign them to a specific tribute or to an entire district. Multiple modifiers on the same tribute multiply together.

Example Modifier	Weight	Effect on Odds
Career Training	×1.5	Tribute is 50% more likely than their score alone suggests
Injured	×0.6	Tribute is 40% less likely
Fan Favourite	×1.2	Slight 20% boost — sponsor advantages
District 2 boost	×1.3	Applies to every tribute in District 2

Note: district modifiers affect everyone in that district. If you want to target one tribute specifically, assign to the tribute, not the district.

Layer B — Alliance Modifier Bleed

20% of the average modifier weight of a tribute's alive allies bleeds into their own modifier. A tribute with no modifier assigned, but allied with someone carrying a ×1.5 boost, will receive a small automatic lift.

Example: Tribute A has no modifier. Their two allies have ×1.4 and ×1.2. Ally average = ×1.3. Tribute A ends up with ×1.06 (20% of the ×0.3 gap).

Layer C — Server Seniority

If a tribute is linked to a Discord server member, their membership length influences their odds. Long-standing members are rewarded; brand-new members take a penalty.

Time in Server	Multiplier	Notes
Less than 30 days	×0.5	New members start as bigger underdogs
30 days – 1 year	×1.0	Neutral — no effect
1–2 years	×1.1	+10% boost
2–3 years	×1.2	+20% boost
3–4 years	×1.3	+30% boost
Each additional year	+×0.1	Continues to scale upward

You can override a tribute's join date using `/admin tribute edit → seniority_date`. Useful for members who rejoined after leaving.

Layer D — District Historical Performance

If your server has historical Games data entered via **/admin history**, each district's past performance subtly shifts their current tributes' odds. This influence is **capped at 15%** — history matters, but it can't override a strong current tribute.

Historical Stat	Effect if Above Average
Kills per Game	Slight boost — this district produces fighters
Average Placement	Boost if low (low = better finish). Inverted stat.
Total Wins (Victors)	Boost — proven winner history
Top-8 Finishes	Boost — consistently deep runs
Top-5 Finishes	Boost — elite performer history
Kill Record	Boost — highest single-game kill total
Runner-Up Finishes/Game	Boost — comes close to winning often
Avg Placement (Last 5)	Boost if low. More weight on recent form.
Bloodbath Kills/Game	Boost — aggressive early-game district

Only stats that have been set contribute. A district with no history set gets a neutral $\times 1.0$ from this layer. Stats are compared to the average across all districts that have data — not an absolute scale.

Stage 4 — Arena Type (Death Cause Only)

This stage only affects **Death Cause** markets. Set the arena type via **/admin game arena** to shift the probability of each cause of death. If no arena type is set, all causes are treated equally.

Death Cause	Artificial Arena	Natural Arena	Explanation
Natural Causes	$\times 0.60$ (less likely)	$\times 1.50$ (more likely)	Exposure, starvation, and hazards are common in wilderness arenas
Mutt	$\times 0.80$ (less likely)	$\times 1.25$ (more likely)	Natural arenas have wildlife and creature threats
Another Tribute	$\times 1.00$ (unchanged)	$\times 1.00$ (unchanged)	Tribute-on-tribute combat is arena-neutral
Gamemakers	$\times 1.50$ (more likely)	$\times 0.60$ (less likely)	Tech traps and constructed hazards dominate artificial arenas

Quick Reference — Admin Controls

Everything below can be changed without touching code. Odds recalculate automatically on the next trigger after any change.

Influence Strength Knobs

Setting	Current Value	Where to Change	What It Does
District peer influence	10%	bot/odds/defaults.py → _DISTRICT_ALPHA	How much alive districtmates shift your odds
Alliance peer influence	20%	bot/odds/defaults.py → _ALLIANCE_ALPHA	How much alive allies shift your odds
Alliance modifier bleed	20%	bot/odds/defaults.py → MODIFIER_ALLIANCE_ALPHA	How much allies' modifiers bleed into yours
District history impact	15%	bot/odds/defaults.py → HIST_ALPHA	Max effect of district historical stats

In-Bot Admin Commands

Command	What It Controls
/admin modifier create/assign	Create and attach custom multipliers to tributes or districts
/admin tribute add/edit	Link a Discord member for seniority; set seniority_date override
/admin game arena	Set arena type (Artificial / Natural) to shift Death Cause odds
/admin history set	Enter district historical stats that feed into Layer D
/admin market odds	Override odds on a specific market — freezes it from recalculation
/admin game start	Opens Pre-Games markets and triggers a full odds recalculation